

D029

Informatics for Transforming Nursing Care

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A nurse informaticist at a hospital with a new rooftop garden reviews a dashboard. A single value shows a patient's heart rate of 128. A trend line shows heart rates on the unit rise before sepsis alerts fire. The nursing informatics team has six members. Which is knowledge?

- A. Recognizing that rising heart rates tend to precede sepsis alerts on the unit.
- B. Deciding to change the sepsis protocol for this patient population.
- C. The six-member team.
- D. The heart rate value of 128.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A nurse sees a patient's potassium of 3.1 mmol/L labeled low on the lab screen. Which DIKW level is this labeled result?

- A. Data.
- B. Knowledge.
- C. Wisdom.
- D. Information.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A clinic sends allergy lists to a hospital through a secure interface built by a well-known vendor. The files arrive intact, but the hospital system cannot tell which entries are drug allergies because each site uses its own local codes. The interface cost \$80,000. What level of interoperability is missing?

- A. Organizational interoperability.
- B. Foundational interoperability.
- C. Semantic interoperability.
- D. None; a more expensive interface would fix it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Two systems use the same terminology codes and message format, but no data-sharing agreement lets the clinic send records. What level is lacking?

- A.** Foundational.
- B.** Organizational.
- C.** Structural.
- D.** Semantic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A quality team at a hospital with a new parking app wants to code hemoglobin A1C lab results consistently across three clinics that use different lab vendors. Which standard fits?

- A.** RxNorm.
- B.** SNOMED CT.
- C.** LOINC.
- D.** ICD-10-CM.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A problem list entry needs a detailed clinical code for “pressure injury, stage 2, sacral region.” Which terminology fits best?

- A.** ICD-10-CM, because it is used for all documentation.
- B.** SNOMED CT.
- C.** RxNorm.
- D.** LOINC.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A sepsis alert fires correctly but appears only in the pharmacy queue six hours after the vital-sign changes. The hospital recently switched to a new pharmacy vendor. What best improves the alert?

- A.** Route it to the bedside nurse and provider at the time vital signs are entered.
- B.** Remove the alert, since CDS never works.
- C.** Add a second sepsis alert.
- D.** Change the pharmacy vendor back.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

On a medical unit with 36 beds, nurses override 95% of drug-interaction alerts, most of them low-severity. A critical alert was recently missed. What should the informatics team do?

- A. Add alerts to all 36 beds' monitors.
- B. Make every alert hard-stop.
- C. Retrain nurses to read all alerts carefully.
- D. Reduce or tier low-severity alerts so high-risk ones stand out.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A hospital's chief nursing officer saw an impressive vendor demo at a national conference and wants to buy a new staffing system. No one has mapped the current scheduling workflow or gathered user needs. What should come first?

- A. Analyze current workflows and user requirements.
- B. Pick the product with the best conference demo.
- C. Sign the contract and plan training.
- D. Go live on one unit to see what happens.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A nurse on a busy cardiac unit who recently won a teaching award looks up a coworker's chart out of concern after hearing she was admitted. The nurse is not involved in the coworker's care. Which HIPAA principle is violated?

- A. No violation, because staff may view any chart in their hospital.
- B. Privacy Rule: access only the minimum necessary for one's job.
- C. Security Rule only, because the chart is electronic.
- D. No violation, since the intent was caring.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A hospital encrypts laptops, requires unique logins, and keeps audit logs of chart access. Which HIPAA rule do these measures address?

- A.** The Breach Notification Rule.
- B.** The Privacy Rule.
- C.** The Security Rule.
- D.** None; HIPAA does not address electronic data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A nurse manager on a unit with a new break room wants monthly reports on how many patients received fall-risk interventions. Nurses document interventions only in free-text notes. What is the main obstacle?

- A.** The report needs more nurses to read notes.
- B.** Fall-risk data can never be reported.
- C.** The break room distracts nurses.
- D.** Free-text notes are unstructured and hard to query; structured fields are needed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A nurse informaticist at a hospital ~~with a new rooftop garden~~ reviews a dashboard. A single value shows a patient's heart rate of 128. A trend line shows heart rates on the unit rise before sepsis alerts fire. ~~The nursing informatics team has six members.~~ Which is knowledge?

Target word: identify knowledge **Decides:** A pattern drawn from information.

A	Correct	A pattern from information is knowledge.
B	Inverted operation	Applying knowledge to act is wisdom.
C	Camouflage	Not part of the DIKW chain.
D	Confusion set	A raw value: data (or information with context).

2. Answer D Concept 1 • transfer

A nurse sees a patient's potassium of 3.1 mmol/L labeled low on the lab screen. Which DIKW level is this labeled result?

Target word: level of a value with context **Decides:** A value given context and meaning.

A	Dropped element	Data would be the bare number without context.
B	Adjacent category	Knowledge is a pattern across information.
C	Inverted operation	Wisdom is acting on knowledge.
D	Correct	Data given context (flagged as low) is information.

3. Answer C Concept 2

A clinic sends allergy lists to a hospital through a secure interface ~~built by a well-known vendor~~. The files arrive intact, but the hospital system cannot tell which entries are drug allergies because each site uses its own local codes. ~~The interface cost \$80,000.~~ What level of interoperability is missing?

Target word: missing level **Decides:** Data move but meaning is not shared.

A	Adjacent category	Policy and governance are not the described gap.
B	Dropped element	Data already arrive intact.
C	Correct	Standard codes are needed so both systems share meaning.
D	Camouflage	Cost does not create shared meaning.

4. Answer B Concept 2 • transfer

Two systems use the same terminology codes and message format, but no data-sharing agreement lets the clinic send records. What level is lacking?

Target word: missing level **Decides:** Governance/policy gap.

A	Adjacent category	The issue is permission, not the technical ability to connect.
B	Correct	Agreements, policy, and workflow are organizational.
C	Confusion set	Format is already shared.
D	Inverted operation	Meaning is already shared by common codes.

5. Answer C Concept 3

A quality team at a hospital with a new parking app wants to code hemoglobin A1C lab results consistently across three clinics that use different lab vendors. Which standard fits?

Target word: terminology for lab tests **Decides:** Lab tests and observations.

A	Inverted operation	RxNorm codes medications.
B	Confusion set	SNOMED codes clinical findings and problems.
C	Correct	LOINC codes lab tests and observations.
D	Adjacent category	Diagnosis codes for billing and statistics.

6. Answer B Concept 3 • transfer

A problem list entry needs a detailed clinical code for “pressure injury, stage 2, sacral region.” Which terminology fits best?

Target word: terminology for a clinical finding **Decides:** Clinical findings on a problem list.

A	Absolute language	ICD-10 is built for billing and statistics, not all documentation.
B	Correct	Designed for detailed clinical findings.
C	Adjacent category	Medications only.
D	Confusion set	Codes lab tests and observations, not the problem itself.

7. Answer A Concept 4

A sepsis alert fires correctly but appears only in the pharmacy queue six hours after the vital-sign changes. ~~The hospital recently switched to a new pharmacy vendor.~~ What best improves the alert?

Target word: CDS fix **Decides:** Wrong person and wrong time.

A	Correct	Right person and right time in the workflow.
B	Absolute language	Absolute and unsupported.
C	Inverted operation	More alerts do not fix delivery.
D	Camouflage	The vendor is not the cause.

8. Answer D Concept 5

On a medical unit ~~with 36 beds~~, nurses override 95% of drug-interaction alerts, most of them low-severity. A critical alert was recently missed. What should the informatics team do?

Target word: fix alert fatigue **Decides:** Too many low-value alerts.

A	Camouflage	Bed count does not decide the fix.
B	Inverted operation	More interruption worsens fatigue.
C	Dropped element	Ignores the system cause.
D	Correct	Fewer, higher-value alerts restore attention.

9. Answer A Concept 6

A hospital's chief nursing officer saw an impressive vendor demo ~~at a national conference~~ and wants to buy a new staffing system. No one has mapped the current scheduling workflow or gathered user needs. What should come first?

Target word: first life-cycle step **Decides:** Analysis of needs and workflow.

A	Correct	Needs analysis comes before selection.
B	Camouflage	A demo is not a needs analysis.
C	Inverted operation	Selection before analysis reverses the order.
D	Dropped element	Skips analysis, design, and testing.

10. Answer B Concept 7

A nurse on a busy cardiac unit ~~who recently won a teaching award~~ looks up a coworker's chart out of concern after hearing she was admitted. The nurse is not involved in the coworker's care. Which HIPAA principle is violated?

Target word: principle violated **Decides:** Access without a work need.

A	Absolute language	Access is limited to job need.
B	Correct	Viewing PHI without a care role violates privacy.
C	Adjacent category	The core issue is an improper use, governed by the Privacy Rule.
D	Inverted operation	Intent does not permit access.

11. Answer C Concept 7 • transfer

A hospital encrypts laptops, requires unique logins, and keeps audit logs of chart access. Which HIPAA rule do these measures address?

Target word: which rule **Decides:** Safeguards for electronic PHI.

A	Adjacent category	Covers what happens after a breach.
B	Confusion set	Governs uses and disclosures of PHI.
C	Correct	It requires administrative, physical, and technical safeguards for ePHI.
D	Inverted operation	The Security Rule is specifically about ePHI.

12. Answer D Concept 8

A nurse manager ~~on a unit with a new break room~~ wants monthly reports on how many patients received fall-risk interventions. Nurses document interventions only in free-text notes. What is the main obstacle?

Target word: obstacle to reporting **Decides:** Unstructured data.

A	Dropped element	Treats a data structure problem as staffing.
B	Absolute language	Structured documentation makes it reportable.
C	Camouflage	Unrelated to reporting.
D	Correct	Coded fields can be counted and reported.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	DIKW hierarchy	Data are raw facts; information is data given context; knowledge is patterns drawn from information; wisdom is applying knowledge to act well.	Calling a single number information or a trend data.	A raw value vs. an interpreted pattern vs. a decision.
2	Levels of interoperability	Foundational = data can move; structural = format is shared; semantic = meaning is shared through standard codes; organizational = policy and workflow support exchange.	Assuming that sending a file means the receiver can understand it.	Data arrive but the receiving system cannot interpret or use them.
3	Standard terminologies	LOINC codes lab tests and observations; SNOMED CT codes clinical findings and problems; ICD-10-CM codes diagnoses for billing; RxNorm codes medications.	Using a billing code set for detailed clinical documentation, or mixing up LOINC and SNOMED.	"Which terminology should be used to code..."
4	Five rights of clinical decision support	Deliver the right information to the right person in the right format through the right channel at the right time in the workflow.	Adding more alerts as the default fix.	CDS that is correct but ignored, late, or sent to the wrong person.
5	Alert fatigue	Too many low-value alerts train users to override all alerts; the fix is fewer, tiered, high-value alerts.	Believing more alerts always increase safety.	High override rates.
6	System life cycle order	Analyze needs and current workflow before selecting or designing a system; then design, build, test, implement, and evaluate.	Buying a product first and fitting workflows to it later.	A project that starts with a vendor demo.
7	HIPAA Privacy vs. Security Rule	The Privacy Rule governs uses and disclosures of all protected health information and sets minimum necessary; the Security Rule requires safeguards for electronic PHI.	Thinking HIPAA covers only electronic records, or that any staff member may view any chart.	A question about who may see data vs. how electronic data are protected.
8	Structured vs. unstructured data	Structured, coded fields can be queried, aggregated, and used by decision support; free text is hard to reuse for analytics.	Assuming data in free-text notes are ready for reports.	A data request that cannot be answered from narrative notes.

Course scope (public description): Informatics for Transforming Nursing Care integrates nursing science with multiple information and analytical sciences to identify, define, manage, and communicate data, information, knowledge, and wisdom in nursing practice. Students will acquire knowledge and skills to apply informatics concepts, communications, and data that are critical to facilitating interprofessional data-driven decision-making. It is designed to build competence in the use of patient- and population-based applications that inform and support the transformation of nursing care delivery toward a future of value-based quality nursing care that improves health outcomes. This course aligns theoretical concepts with practical applications of informatics and is consistent with the functional areas and responsibilities of informatics nurses as defined by the American Nurses Association

Scope and Standards for nursing informatics.